

Research and Market Agreement Across Frontier Models: An Expanded Prospective Comparison

Vorhersage experimental series

September 18, 2026 (America/New_York)

Abstract

We compare three language models on 15 unresolved prediction-market contracts, using identical questions and a common set of outside evidence. Each model produces a question-only forecast and a separate research-informed forecast, for 90 calls in total. Market prices are committed before research and withheld from all forecasting inputs. On the 13 contracts with accepted research forecasts from every model, mean absolute distance from the opening Kalshi midpoint is 8.92 percentage points for Claude Fable 5.1, 11.08 for GPT-6 Astra, and 13.08 for Gemini 3.1 Pro. Equal-weight averaging yields 10.87 points. On the nine contracts with all six accepted forecasts, research reduces mean distance by 38–45% across models. An overly broad lexical screen excludes ten otherwise valid responses; an explicitly post-hoc analysis of the original responses preserves the aggregate direction. The results support further study of research augmentation while revealing substantial sensitivity to question composition and response-screening policy.

1 Objective and experimental design

Market probabilities offer a useful intermediate target for evaluating forecasts before events resolve. They incorporate contemporaneous information and permit rapid comparison without importing realized outcomes into the exercise. This study asks whether supplying a fixed body of nonmarket research moves model forecasts toward those contemporaneous prices. It extends an earlier six-contract model comparison with additional questions; results below concern only the new cohort.

The experiment crosses three models with two information conditions. In the *question-only* condition, the worker receives the event definition and forecasting instructions. In the *outside-research* condition, it also receives a frozen evidence packet. Calls are separate: the research worker does not see the question-only response. Thus, a line connecting the two estimates represents a comparison between calls rather than a conversational revision. There is one call per model–question–condition cell. Models share the same prompts and source material within each condition, but retain provider-specific reasoning settings. The design compares the use of common evidence; it does not compare models’ autonomous search strategies or isolate the effect of every stage in the broader forecasting package.

1.1 Selection and chronology

The candidate inventory contained 20 nominated contracts. Before collecting quotes, the protocol fixed that inventory, its order, a no-replacement rule, and eligibility requirements: an active,

nonextreme, uncrossed book with spread at most 10 percentage points and at least one contract available on each side; the outcome must remain unknown, with at least 12 hours before the relevant future event. Three candidates lacked a usable two-sided book: September SpaceX launches, Sam Altman for the Nobel Peace Prize, and *I Love Boosters* for Best Picture.

Two more were excluded before inference. The current annual SpaceX launch total could not be established from the accessible primary material, so an already crossed threshold could not be ruled out. The earthquake definition did not unambiguously state the start of the counting window. The final cohort comprised eight macroeconomic contracts, one major-hurricane count, and six NFL games. No excluded candidate was replaced.

Opening targets were captured around 01:01–01:02 UTC on September 19 (September 18 in New York). Research followed, and the 90 trials were registered at 01:06:45 UTC. All batches were sealed before target revelation at approximately 01:24 UTC. The largest absolute change between opening and follow-up midpoints was 0.5 percentage points. This ordering makes the opening midpoint the fixed primary target; later quotes provide a timing sensitivity rather than a replacement chosen after observing performance.

1.2 Evidence and blinding

Research used primary material from BEA, BLS, the Federal Reserve, the Atlanta Fed, the National Hurricane Center, and NFL.com. Macroeconomic packets contained recent releases and relevant projections, with distinctions between historical measurements, current-quarter nowcasts, and annual projections. NFL packets contained early-season standings and scoring, scheduled games, and injury reports. They were deliberately bounded packets, not comprehensive estimates of team strength. Hurricane evidence relied on captured search-index material from primary NHC pages because direct access returned HTTP 403; this limitation was supplied to the workers.

Kalshi provided event definitions and separately stored evaluator targets. It provided no research evidence or numerical prices to forecasting workers. Workers had no retrieval tools. For each of the 90 responses, the returned system and user prompt receipts, scenario identity, and model identity were checked against the registered inputs. Source packets, jobs, raw results, accepted forecasts, and evaluator targets were preserved with content hashes. The coordinator followed procedural separation of the price store; this was not independent operating-system isolation of that coordinator.

2 Measures and matched comparisons

Let p_{imc} denote the probability from model m for question i under condition c , and let q_i be the committed opening midpoint. We report

$$\text{MAE}_{mc} = \frac{100}{|S|} \sum_{i \in S} |p_{imc} - q_i|, \quad \text{RMSE}_{mc} = 100 \sqrt{\frac{1}{|S|} \sum_{i \in S} (p_{imc} - q_i)^2}.$$

Both are in percentage points. For a given pair, the absolute improvement is $100\{|p_{imQ} - q_i| - |p_{imR} - q_i|\}$. Its percentage improvement divides this quantity by the question-only absolute gap. A zero initial gap has undefined relative improvement; negative improvement denotes deterioration. Aggregate relative improvements below use the ratio of aggregate MAEs, rather than an average of question-level ratios.

Comparisons use common question sets. The primary research-model comparison requires an accepted research forecast from all three models ($n = 13$). The paired research-effect comparison requires all six accepted forecasts ($n = 9$). Available-case scores are also retained in the machine-readable results, but are not used to rank models across unequal question sets. An equal-weight average of the three research probabilities is evaluated on the same 13 questions as its members.

Table 1: Agreement with the opening midpoint: common research cohort ($n = 13$).

Model or benchmark	MAE (pp)	RMSE (pp)
Claude Fable 5.1	8.92	11.28
GPT-6 Astra	11.08	13.84
Gemini 3.1 Pro	13.08	15.64
Equal-weight research average	10.87	12.68
Constant 50%	19.38	22.41

Table 2: Research comparison: fully matched cohort ($n = 9$). Distances and absolute reductions are in percentage points.

Model	Question only	Research	Reduction	Relative
GPT-6 Astra	14.56	8.00	6.56	45.0%
Claude Fable 5.1	17.33	9.56	7.78	44.9%
Gemini 3.1 Pro	15.33	9.44	5.89	38.4%

Research improves aggregate agreement for each model, but individual questions show both gains and losses. For example, all three research forecasts for Q4 GDP are 42%, below the 55% midpoint and farther from it than their question-only forecasts. All three research forecasts for the December Fed decision are 55%, compared with a 30.5% midpoint. Averaging cannot remove an error shared by all members. Conversely, research moves the hurricane forecasts close to the 5% midpoint. The appended question-faceted slope graphs display each accepted estimate and its distance reduction, with a labeled horizontal reference at the midpoint.

The overall model ranking depends on the question mix. Removing NFL contracts from the common research cohort leaves eight questions: Astra’s MAE is 8.44 pp, Gemini’s 9.81 pp, and Claude’s 10.44 pp. NFL contracts additionally have a payoff distinction: a tie pays half, whereas the requested forecast is the probability of a win. Under a simple expected-payout interpretation, the price corresponds to $P(\text{win}) + \frac{1}{2}P(\text{tie})$, not exactly $P(\text{win})$. The non-NFL analysis was specified before inference.

3 Response screening and sensitivity analysis

All 90 calls completed without provider-reported exceptions. The registered acceptance rule nevertheless excluded ten responses, leaving 25/30 for Astra, 25/30 for Claude, and 30/30 for Gemini. All ten contain valid JSON probabilities and valid supplied evidence references. Nine merely deny using market or sportsbook odds. The remaining response mentions the Kalshi naming convention when inferring a home team. None of these ten reports numerical market odds; the lexical matches do not establish price exposure.

The registered exclusions remain unchanged. To expose their effect, a separate *post-hoc* diagnostic includes all 90 original probabilities, without editing text, issuing the excluded forecasts, or obtaining replacement calls. The content review occurred after price revelation and is therefore not part of the preregistered primary analysis.

Table 3: Post-hoc diagnostic using all original responses ($n = 15$). MAE in percentage points.

Model	Question only	Research
GPT-6 Astra	15.67	11.73
Claude Fable 5.1	15.20	9.47
Gemini 3.1 Pro	17.60	13.13

The equal-weight research average in this diagnostic scores 11.31 pp. The aggregate direction and overall research ranking agree with the registered analysis. The screening problem still matters: a future protocol should define how to distinguish explicit denials from assertions of price use before any responses are examined. A vocabulary match is useful as an audit flag but is an imprecise exclusion criterion.

4 Interpretation and next experiments

The study provides additional evidence that the supplied research improves agreement with contemporaneous market assessments on this selected cohort. The comparison is useful precisely because the events remain unresolved: realized outcomes cannot explain the observed agreement. Market disagreement also identifies concrete questions on which evidence selection or numerical interpretation deserves further examination.

The scope of inference remains the observed questions and settings. A two-sided quote screen establishes a usable benchmark, not that every contract has deep liquidity. Related economic releases share information and risks; six games also do not supply six independent replications of a general forecasting method. With one draw per cell and a small number of broader topic groups, we report descriptive differences rather than unsupported precision for a population ranking. These market-distance measures are intermediate outcomes; subsequent realized outcomes can support a separate proper-scoring evaluation.

No transformation $f(p)$ or ensemble weights were fitted. Training one on these same observations and reporting the resulting fit would not establish useful out-of-sample calibration. The next experiment should freeze any proposed correction before testing on later, untouched event groups, preserve a simple identity correction and equal-weight ensemble as controls, and expand coverage across dates and domains. Research packets and response-screening rules should be versioned as experimental factors rather than changed within a completed comparison.

A Package implementation and reproducibility

Vorhersage supplies the deterministic records and evaluation machinery around external research and model judgment. The market workbench stores a question version and contract definition, checks quote eligibility, and places the opening quote in a separate evaluator store. A public hash commits to that target before the research begins. Evidence packets preserve claims, values, source

excerpts, retrieval timestamps, captured content, and declared limitations.

First-class experimental arms combine a method, a model configuration, and a data specification. Here, six arms cross the three models with either no current evidence or the same frozen packet for each question. Registration pins the question versions, information cutoff, arm identities, and one repetition. The transport exports the exact tasks to independent provider calls. The importer checks response identity, JSON structure, references, and the declared acceptance policy before issuing immutable forecast artifacts. Every attempt, including an excluded response, remains in the usage record.

The reveal gate requires terminal records for every registered trial and seals for all model batches. It verifies raw-result and forecast commitments before opening prices. The published comparison then scores all accepted model/condition forecasts against each question’s committed target. A representative forecast in the workbench journal is only the bridge to that reveal; it is not the source of the multi-model comparison. An offline verifier independently recomputes MAE, RMSE, and distance outside the bid–ask spread, and checks target hashes, retrieval-before-registration timing, and seal-before-reveal ordering.

Table 4: Reported provider inference cost, including excluded responses.

Model	Calls	USD
GPT-6 Astra	30	0.99960
Claude Fable 5.1	30	2.25114
Gemini 3.1 Pro	30	0.39297
Total	90	3.64371

Exact identifiers are `gpt-6-astra`, `claude-fable-5-1`, and `gemini-3.1-pro-preview`. Astra uses high reasoning effort; Claude uses adaptive thinking with high output effort; Gemini uses a 2,048-token thinking budget and temperature 0.5. All have an 8,192-token configured output limit. Serialized legacy sampling defaults may be omitted or ignored by adapters for reasoning models; the archived configurations and raw provider responses are the operational record. These settings are not equal compute budgets. The reported costs exclude research/coordinator work, for which cost is unknown.

Artifacts reside in `examples/kalshi_expansion_20260918/`. The accompanying README maps protocols, evidence, tasks, jobs, results, seals, and target disclosures. `run/comparison.json` contains all cohort metrics; `run/output-review-sensitivity.json` contains the post-hoc review and diagnostic. The saved database integrity check passed for 415 artifacts. The targeted study tests pass 30 tests. No core package dependency was added for this expansion.

The following commands verify the saved study and regenerate its figures and PDF from the repository root (with the study dependencies installed):

```
.venv/bin/python examples/kalshi_expansion_20260918/verify_results.py
Rscript examples/kalshi_expansion_20260918/plot_performance.R
latexmk -cd -pdf examples/kalshi_expansion_20260918/report.tex
```

The offline check needs neither inference calls nor private-store access. The following ggplot2 plates show the primary comparison: blue circles denote question-only forecasts, green squares research forecasts, and dashed lines the opening midpoint. Missing responses are never assigned zero probability.

New contracts, shared evidence: part 1 of 3

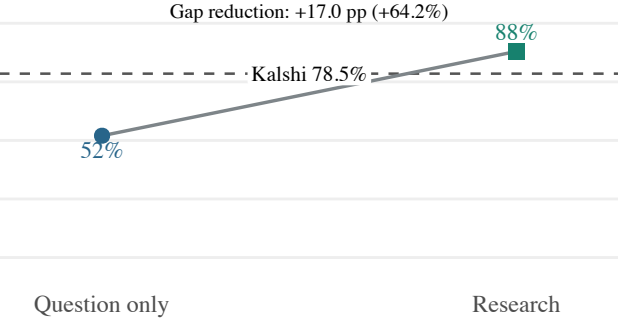
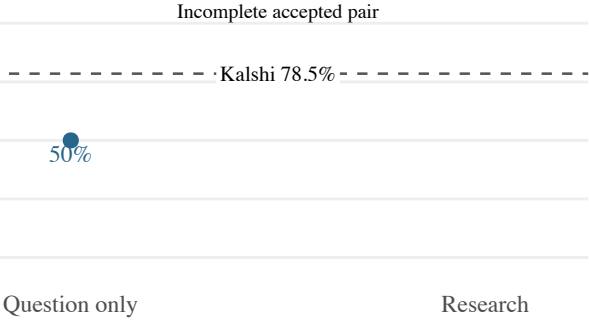
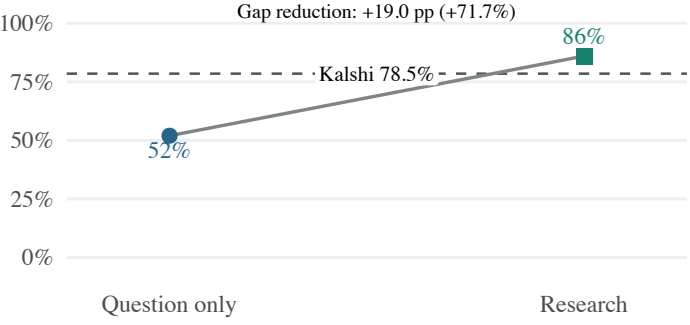
Question-only □ outside research; dashed line: sealed opening midpoint

GPT-6 Astra

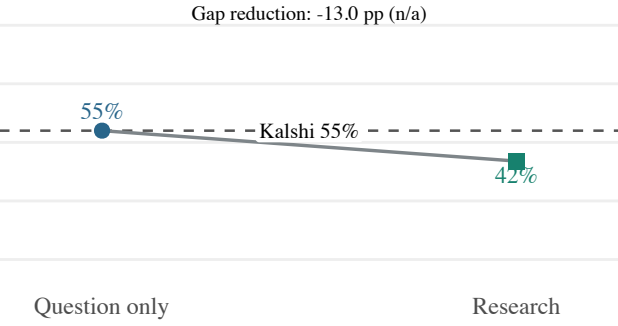
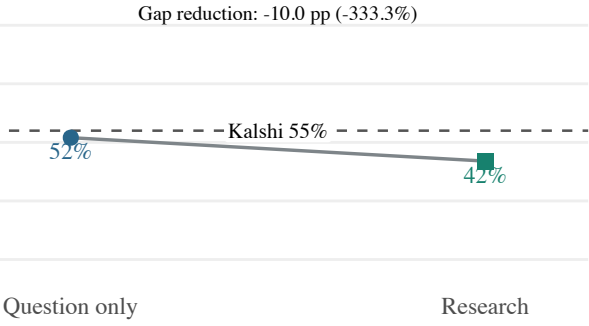
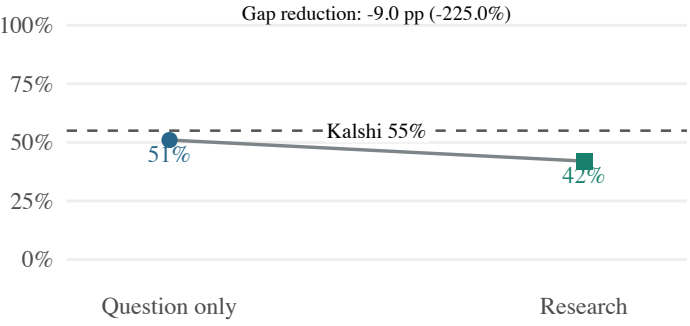
Claude Fable 5.1

Gemini 3.1 Pro

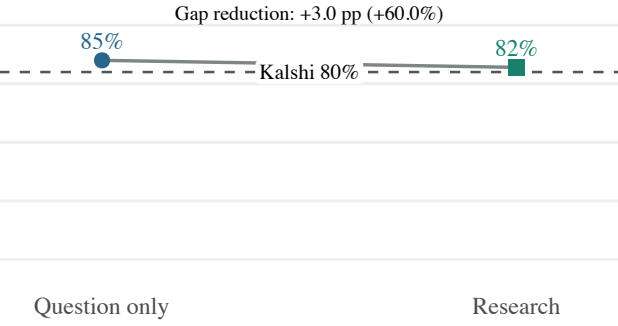
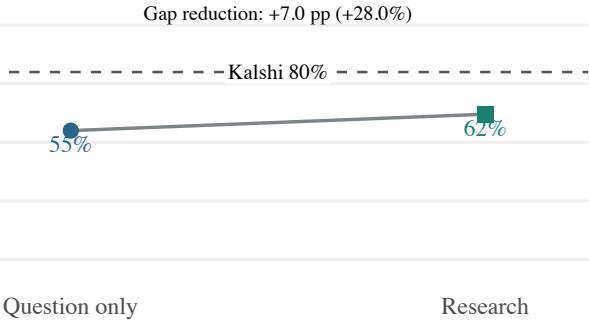
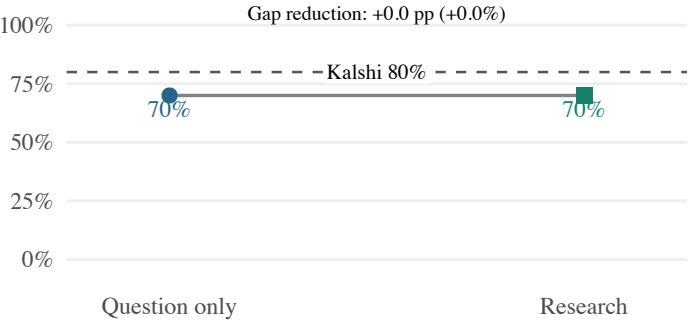
Will Q3 2026 real GDP growth exceed 2% annualized?



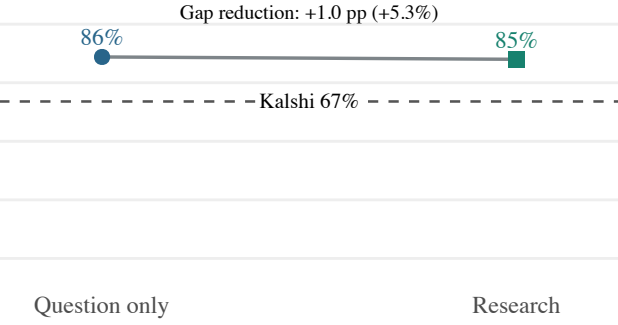
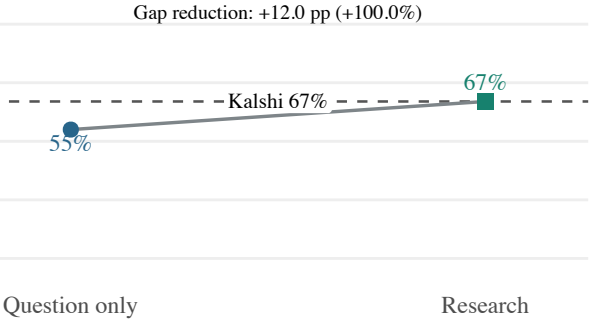
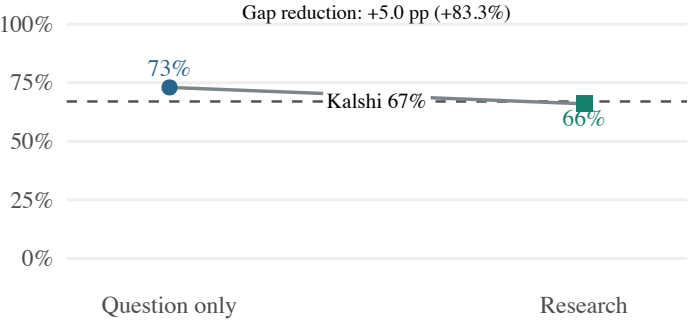
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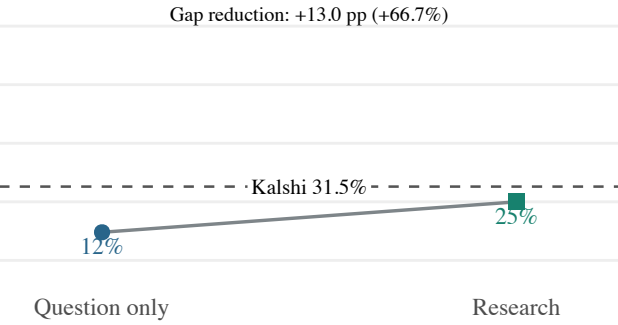
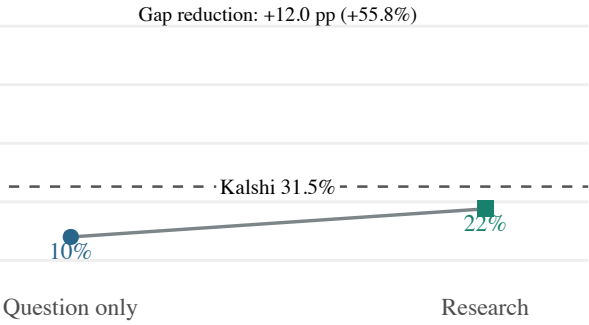
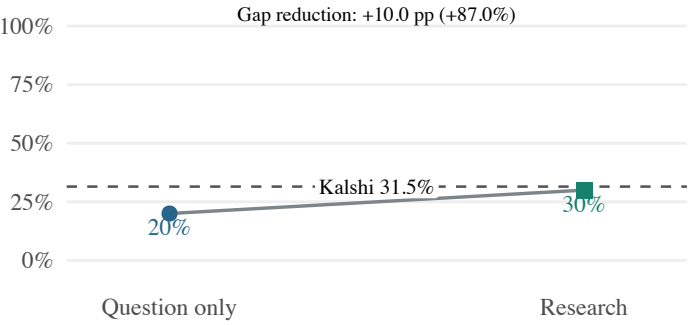
Will September payroll growth exceed 30,000?



Will October payroll growth exceed 30,000?



Will October monthly CPI growth exceed 0.4%?



New contracts, shared evidence: part 2 of 3

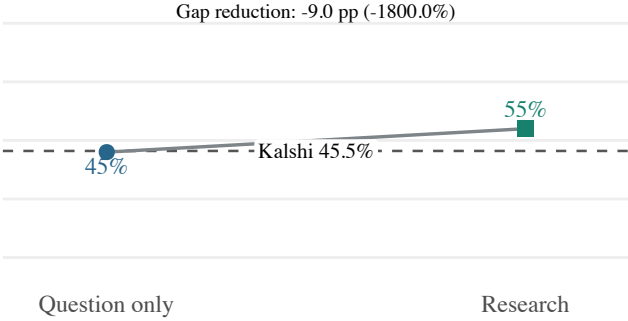
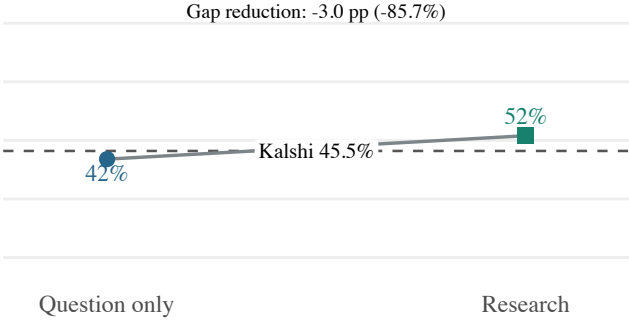
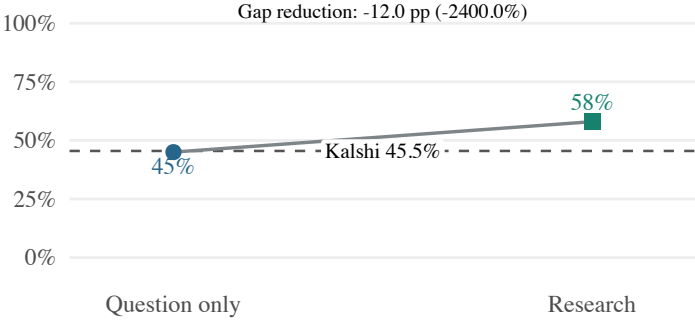
Question-only ☐ outside research; dashed line: sealed opening midpoint

GPT-6 Astra

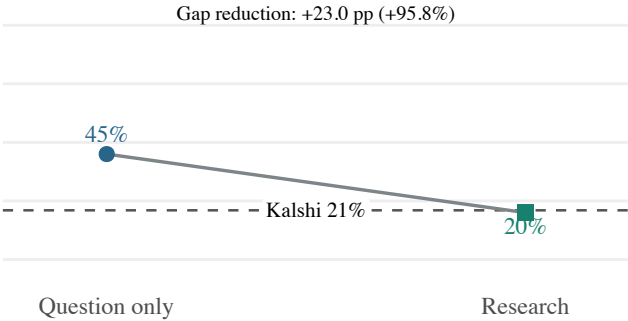
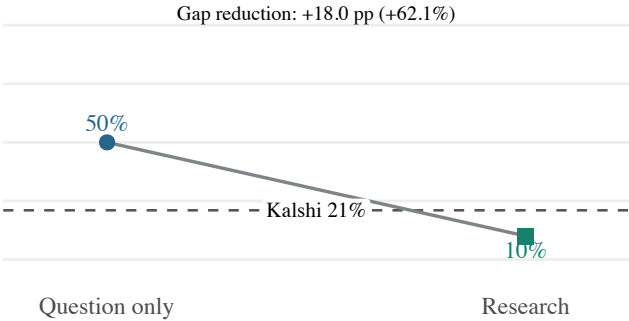
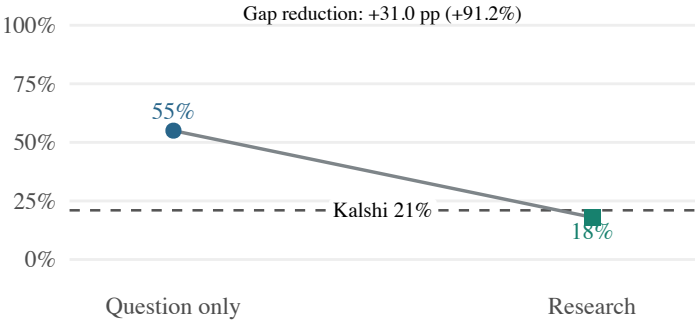
Claude Fable 5.1

Gemini 3.1 Pro

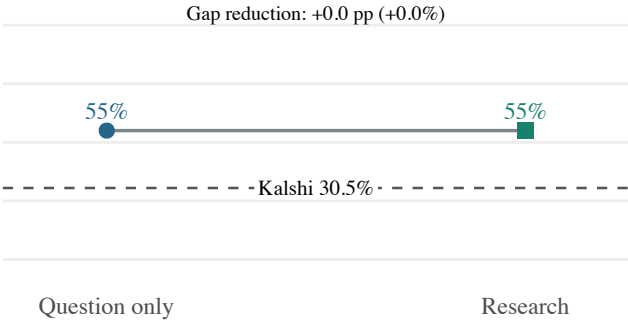
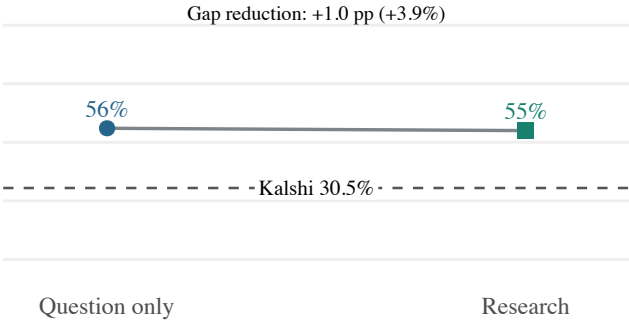
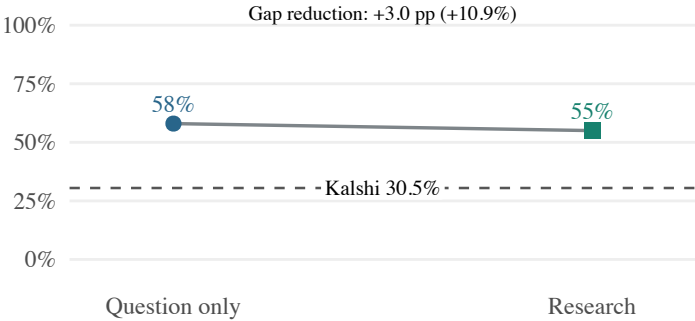
Will November monthly CPI growth exceed 0.2%?



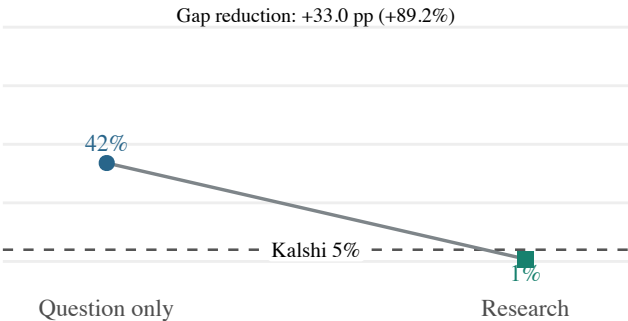
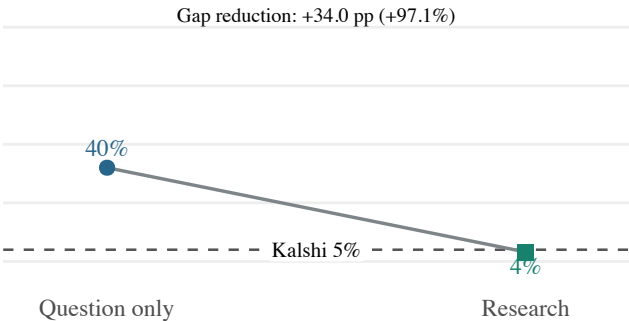
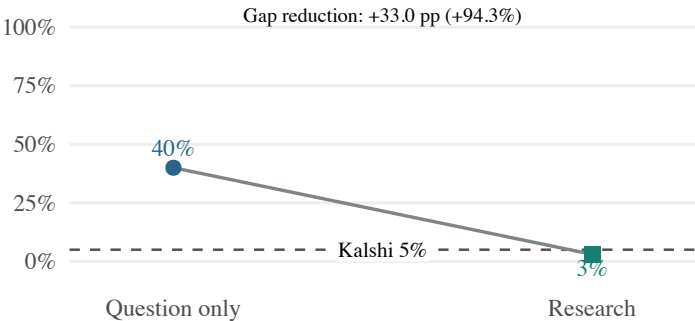
Will October unemployment exceed 4.3%?



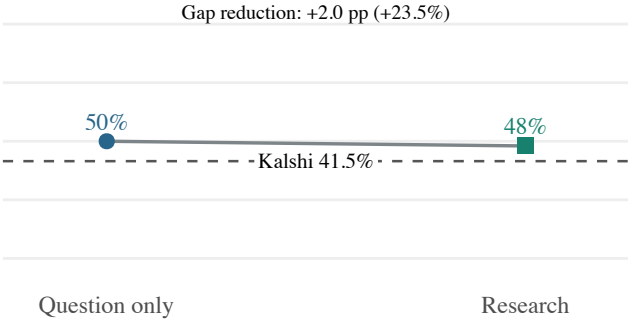
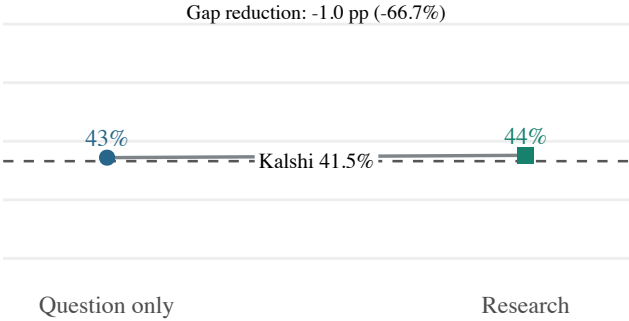
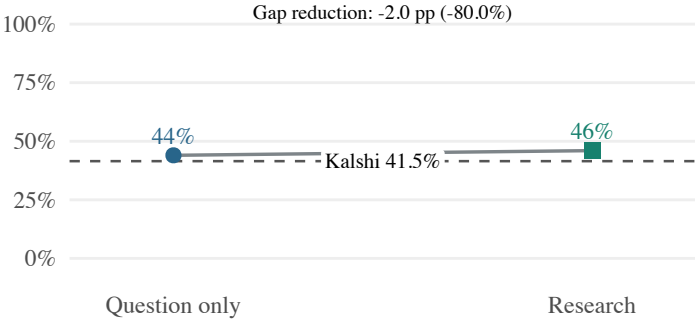
Will the Fed hold rates at its December meeting?



Will 2026 Atlantic major hurricanes exceed three?



Will Cincinnati beat Houston on September 20?



New contracts, shared evidence: part 3 of 3

Question-only ☐ outside research; dashed line: sealed opening midpoint

Forecast probability



Registered acceptance rule; excluded responses are omitted, never set to zero. One draw per cell.
Gap reduction is the decrease in absolute distance from the midpoint; negative values indicate deterioration.
NFL ties have fractional payouts. Related economic releases and games are not independent observations.